

Compass Hill Estate — Master Property Plan (Southwest Ohio)

Revision B · September 30, 2026. Supersedes the "Berm Ring Estate" draft. The estate is now **Compass Hill**. "Berm Ring" stays the name of the Zen garden and storm basin south of the house.

Compass Hill can be built as designed. Checking it against current code turned up five issues, and each one has a fix:

- **NiFe battery bank:** NEC 480.3 requires non-lead-acid batteries to be listed. The bank therefore needs written approval, or a field label from a nationally recognized testing laboratory (NRTL).
- **Drive width:** the S-bend may have to be 26 ft if the fire official measures the dead end as longer than 500 ft.
- **Hose reach:** the south corners of the walkout need either a sprinkler-based extension of the 150-ft hose rule or an emergency lane on the east side.
- **Berm Ring sizing:** it should hold a 6.84-in, 24-hour storm.
- **Budget:** plan with no federal clean-energy tax credits.

What Revision B adds. It commits the decisions that were still open (§22.1), lists what is waiting on the land, the energy model, or officials (§17), fills in items that were never specified, corrects the parts of the draft that were missing or wrong, and adds the cart-and-utility tunnel system (§13). Appendix A maps every revision item to the section where it now lives.

TL;DR

- **Build as designed.** As of September 2026, Ohio still permits houses under the 2019 Residential Code of Ohio (RCO, based on the 2018 IRC). The five-year review is open for comment until Sept. 30, 2026, and it keeps the 2019 RCO as its base. RCO R327 requires UL 9540 listing, and NEC 480.3 requires listing for all non-lead-acid batteries. So the LFP bank goes in during Phase 1, and the NiFe bank waits for written approval or an NRTL field label.
- **Size the Berm Ring for 6.84 in of rain in 24 hours.** NOAA Atlas 14 for Lebanon gives a 100-year, 24-hour storm of 5.70 in (upper 90% bound 6.09 in). The design rule takes the larger of that upper bound and the 5.70-in point value plus 20%, which is 6.84 in. That needs about 11,000 cu ft of storage below the line 12 in under the walkout sills. The ring impounds about 0.25 acre-ft, far below Ohio's 15-acre-ft dam-safety exemption; get the exemption confirmed in writing.
- **Tunnels tie the hilltop together.** A cart-and-utility loop wraps the cistern under the pavilion. Tunnels run from it to the house and the garage now, and to the gate later. Every utility past the meter runs in them. The loop has to be built with the cistern, so it lands in Phase 1.
- **Get the fire-water number early.** Without hydrants, the fire official may size the fire reserve by building volume. My rough estimate is 13,000–32,000 gal, which could take most or all of a 30,000-gal cistern. Ask while you're checking out land, and plan for about 40,000 gal if needed (§3.6).
- **Budget with no tax credits, and phase the work.** The §25D credit ended for expenditures after Dec. 31, 2025. For new construction the IRS counts the expenditure as made when you start using the house, so this house cannot qualify. My own rough placeholder, not a quote, is now about **\$19–35M**: \$16–28M for the estate plus \$3.2–7.3M for the tunnels. House width is the biggest cost dial: each 10 ft adds or removes 1,440 sq ft. The budget ceiling is set after the first professional estimate.

Tags used throughout:

- **[F]** = FINAL decision
- **[R]** = RECOMMENDATION adopted by default (also listed in §22)
- **[A]** = ASSUMPTION to verify
- **Δ** = code or safety conflict, always paired with a fix

1. Executive Summary and Governing Values

A single north-south axis runs from a rural road near Lebanon, Warren County (about 39.4°N), down a south-facing hill. In order along it:

1. A hidden truck lay-by and a gated entrance court.
2. An S-bend drive.
3. A hilltop motor court with a slate-coned pavilion standing on the fire cistern. The cart-and-utility tunnel loop wraps the cistern below.
4. A porte-cochère at the front door.
5. A 120 × 48-ft, three-level ICF house at the brow. It is earth-sheltered on the north and looks south over the "Berm Ring" Zen garden, which doubles as the storm basin.
6. Clean-water outfalls feeding a seasonal pool and meadow.
7. Legacy oak woods in the lower corners, protected by a conservation easement.

The garage and its battery basement sit off the axis, to the west. Tunnels link the house, the court, and the garage below grade now, and the gate later.

Value	Meaning	Where it shows
Enduring (top priority; tiebreaker)	Materials that last 100+ years, standard parts, full records	S1 slate, stainless, ICF, limestone, parts library, house book, serviceable tunnels
Passive	Sun, shade, earth, and gravity first; machines as backup	Overhangs tuned to 39.4°N, berms, thermal tank, gravity drainage
Regenerative	Net-positive energy; water leaves cleaner than it arrived	Two outfalls, no-chloride covenant, easement, legacy woods
Majestic	Grandeur through setting and proportion	Hidden-valley reveal, the axis, the pavilion cone
Natural	Looks as if it grew from the hill	Excavated limestone, site-grown white oak, native plants

Do these first:

1. Hire a soils (geotechnical) engineer and a civil engineer now to screen parcels.
2. Hold pre-design meetings with the building and fire officials covering the battery vaults, gate, dry hydrant, fire-water reserve, access length, hose reach, tunnels, and code edition.
3. Once you own a parcel, hire an architect and an energy modeler, and commission the energy model.
4. Get a first professional cost estimate. Then set the budget ceiling and confirm the house width and ICF scope.
5. Set the code-timing strategy (§22).
6. Get insurance quotes on the battery vaults before committing to nickel-iron.

2. Parcel, Hill, Search, Due Diligence

Parcel criteria [F]:

- **Size:** 25–30 acres. The plan uses about 13; the illustrative parcel is 900 × 1,300 ft, about 27 acres.
- **Slope:** faces south at 8–12%, ideally 8–10% across the 48-ft house depth.
- **Hilltop:** a flat-ish area of at least 250 × 180 ft, plus room for the lay-by.
- **House contour:** about 250 ft wide at the brow.
- **Road:** on the north side, with at least 400 ft of frontage.
- **Downhill land:** owned, for the spreaders, pool, and meadow, and well above creeks and floodplain.
- **Disqualifiers:** springs, seeps, or signs of past slides.

Where to look [F]:

- **First choice:** the south-facing north rims of east–west valleys in eastern Warren, Clinton, and southern Greene counties. That means the Little Miami, Todd Fork, and Caesar Creek valleys, near Caesar Creek State Park and Fort Ancient.
- **Next:** Brown, Highland, and Adams counties.
- **Avoid:** flat west-central and northwest Ohio.
- **Avoid:** Cincinnati-area river hills on Kope Formation colluvium (the loose soil that collects on those slopes).
 - ODNR describes the Kope as "a 220-foot-thick, shale-rich unit" and says it slumps and flows when wet.
 - ODNR also cites a USGS finding that Cincinnati has "the highest per capita cost from landslide damage of any city in the United States."
 - A LiDAR-based inventory mapped 842 landslides in the area.
- **Rule:** reject any site with Eden or Fairmount soils, Kope colluvium under the footprint, or slopes over 20% within the building envelope.

Due diligence [F]:

Task	Detail / fix
Web Soil Survey + LiDAR	Look for scarps, hummocks, and colluvial aprons
County auditor GIS	Property lines, flood zones, CAUV status, easements
Site walks	Late winter, and again after heavy rain
Geotech borings + slope stability	Do them in the same site visit as the health district's septic soil evaluation
Survey	1-ft contours, which the Berm Ring storage check needs
Zoning	Confirm front-yard accessory structures (gate buildings, berms) are allowed
Utilities	Public water or a well; sewer or septic type; a natural-gas main along the road or propane; fiber at the road (Starlink is the backup either way). See §17.1
Fire water	Ask the fire official for the required fire-water reserve and whether sprinklers earn a reduction (§3.6)

Task	Detail / fix
Oak and limestone yield	Have a forester cruise the trees inside the building area (the only oak that can be cut, \$14) and have the geotech estimate usable limestone from the borings
CAUV recoupment	ORC 5713.34 charges "the tax savings on the converted land during the three tax years immediately preceding," recorded as a lien. Fix: have the seller pay it or credit it in the purchase agreement, and keep unbuilt acres in qualifying use
⚠ CAUV and the easement	CAUV counts woodland only when it is devoted to commercial timber production, and the easement bans harvest. Fix: before closing, ask the county auditor which acres can stay in CAUV once eased [A], and have the seller's credit cover every acre that will leave CAUV, not just the building area (\$14)
Mineral rights / old mines	Check only if searching farther east
Bat timing	ODNR/USFWS guidance recommends tree clearing Oct 1–Mar 31 and discourages bedrock work Nov 15–Mar 31 near hibernacula. Fix: clear in winter; drill the geothermal bores April–October

3. Master Site Plan

3.1 Zones, road to south line [F]

1. **Road frontage:** 15-ft sight strip, planted separator berm, one-way lay-by, evergreen belt behind.
2. **Entrance court:** stem and visitor pads, 96-ft teardrop, gate, two gate buildings.
3. **S-bend drive.** The gate tunnel follows it underground later.
4. **Hilltop:**
 - Motor court with the pavilion and cistern at its center, and the tunnel loop wrapped around the cistern below.
 - Forecourt, drop-off loop, and porte-cochère.
 - Garage loop to the west, with the garage tunnel's covered ramp.
 - Utility transformer beside the court.
 - Septic field and reserve under meadow on the east shoulder.
 - Well on the west side, with the drinking-water and sprinkler tanks buried beside it.
5. **Brow:** the house, porches, and corner decks.
6. **Berm Ring:** Zen garden, rain-garden basin, and stone arch outlet.
7. **Mid-slope:** clean-water spreader, seasonal pool, meadow and view corridor with open-grown oaks, and the working spreader at a woods edge.
8. **Lower corners and side strips:** legacy woods and trails.
9. **South line:** a low belt of eastern red cedar.

3.2 Site diagram (illustrative 900 × 1,300 ft; north at top; distances from the road)

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PUBLIC ROAD (north) ===== 0'
mailbox E of exit | 15' mowed sight strip | separator berm ~22' base
<-entry(W) \_____ ONE-WAY LAY-BY 28' x ~315' _____/ exit(E)-> ~45-73'
      [semi 75']_____ | STEM 24'x57' | _____[semi 75']
berm (wildlife break W) [pad][pad] [pad][pad] 20x18 ea ~130'
      .-- TEARDROP 96' OD, ~40' island --.
      '-----+-----' ~226'
[CARRIAGE HOUSE 26x60] ===GATE 24'=== [GATEHOUSE 26x60]
      : gate-end chamber + ramp (later)
      | S-bend 24'* (~250') ~266-516'
      : gate tunnel below (later)
GARAGE 48x30 <-- west loop --( MOTOR COURT 150' OD )--> E shoulder:
(2 vaults below; PAVILION on CISTERN SEPTIC+RESERVE
garage lobby + ramp) tunnel LOOP below; 7 pads ~516-666'
      :.... garage tunnel ....: transformer beside court
      FORECOURT -> DROP-OFF -> PORTE-COCHERE ~666-736'
well (W) +----- HOUSE 120' x 48' -----+ ~736-794'
+ buried [20x20] 6' dry-under deck over walkout [20x20]
water tanks BERM RING ( yard 140x70 ~> brook ~> BASIN 140x25 ) ~794-909'
      \___ STONE ARCH 8x8 + spillway ___/
SW woods CLEAN SPREADER (60-100' lip) ~960'
(working SEASONAL POOL + low-flow bypass (no fish) ~1,020'
spreader, MEADOW / VIEW CORRIDOR, 3-4 open oaks SE woods
>=200' from pool)
===== 1,300'
evergreen belts on both side lines; low red cedar on south line
: = tunnel (below grade; see Section 13 for the hilltop layout)
* 26' if the fire official measures the dead end >500' (3.5)

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3.3 Site element schedule

Element	Location	Purpose	Dimensions / materials
Sight strip + berm	Road edge	Sight distance; screening	15 ft + ~22-ft berm base; compacted fill
Truck lay-by	Behind berm	Pull-through semi parking	28 × ~315 ft; heavy-duty concrete for 80,000-lb trucks
Stem + 4 visitor pads	Middle of lay-by	Drop-offs, visitors	Stem 24 × 57 ft; pads 20 × 18 ft
Teardrop	End of stem	Fire and truck turnaround	96-ft outer diameter, ~40-ft island, cobble apron
Gate	South end of teardrop	Security line	24-ft cantilever; UL 325 operator
Gate buildings	Straddling the gate	Check-in and equipment (east); porch and carts (west)	~26 × 60 ft each; limestone, S1 slate
S-bend drive	Gate to court	Concealed approach	24 ft wide (26 if required) × ~250 ft
Motor court	Hilltop	Arrival; fire turnaround	150-ft outer diameter, 24-ft ring
Pavilion + cistern	Court center	Gathering; fire and non-potable water	Pavilion up to 50 ft across; cistern 30-40 ft across, 30,000 gal (40,000 if the fire reserve requires it)
Covered pads (7)	Court spokes	Covered parking	Two-car each; oak frames, hipped slate; 4 built in Phase 1

Element	Location	Purpose	Dimensions / materials
Walk + porte-cochère	Court to front door	Shelter	12 ft wide, 14 ft clear; slate; lit ceilings; car-stopping posts
Garage + loop	West of court	Cars, PV, batteries	48 × 30 ft; floor level with the house main floor
Tunnel loop + chambers	Under the court	Cart turnaround; utility spine; cistern valves	Loop wraps the cistern; six ~24 × 24-ft chambers (\$13)
Tunnels	Court to house, garage, and gate	Carts and all utilities past the meter	Main and garage tunnels in Phase 1; gate tunnel later
Cart ramps (2)	Garage loop; just inside the gate (later)	Cart entrances	~90-110 ft each, roofed, drained top and bottom
Utility transformer	Beside the court	Service entrance	Utility's own line in its own conduit
House	Brow	Residence	120 × 48 ft, three levels; ICF
Berm Ring	South of house	Storm storage; garden	~7 ft tall; yard 140 × 70 ft
Arch + spillway	South berm	Outlet; emergency exit route	~8 × 8 ft; limestone
Clean spreader + pool	Mid-slope	Clean outfall; habitat	60-100-ft level stone lip; fishless seasonal pool
Working spreader	SE or SW woods edge	Treats pavement runoff	40-60-ft lip + vegetated filter strip
Septic + reserve	East shoulder	Wastewater	Mound, drip, or aerobic system (set by the soil evaluation)
Well	West side	Drinking water	Setbacks per OAC 3701-28 [A]; line enters the garage tunnel through a short sleeve
Drinking-water + sprinkler tanks	Buried beside the well	Potable storage; sprinkler supply	1,000-2,000 gal potable; sprinkler tank sized by the designer
Fuel	Gas main, or a buried propane tank	Generator (and a pavilion grill, if wanted)	1,000-gal buried tank if there's no main; siting in §9
Trails	Woods and meadow	Carts, walking	8 ft wide; crushed limestone

3.4 Area tabulation (planning estimate, 27-acre parcel)

Zone	Acres
Entrance complex	1.6
S-bend drive	0.3
Motor court + pavilion	0.6
Forecourt, drop-off, porte-cochère, walk	0.3
Garage + loop	0.4
House, porches, decks	0.2
Berm Ring	0.6
Septic + reserve	1.5
Well area	0.2
Creeks, spreaders, filter strip	0.8

Zone	Acres
Seasonal pool + buffer	0.8
Meadow / view corridor	4.0
Trails	0.7
Lawns, grading margins	1.0
Developed subtotal	≈13.0
Privacy belts	≈3.2
Legacy woods	≈10.8
Total	≈27.0

The tunnels run under the court, forecourt, garage loop, and S-bend, so they add no surface area. The project disturbs more than 1 acre (impervious surface alone is about 3.2 acres), so Ohio EPA's Construction General Permit (CGP) applies.

3.5 Fire-access checks

- **Turnarounds.** IFC Appendix D, Table D103.4, accepts a 96-ft-diameter cul-de-sac for dead ends of 151–500 ft. The teardrop and the court ring both meet this [F].
- **△ Drive width.** The same table requires 26 ft of width for dead ends of 501–750 ft. The illustrative road-to-court distance is about 520 ft.
 - **Fix [R]:** confirm with the fire official whether Appendix D is enforced and how the length is measured.
 - If it counts as over 500 ft, either widen the S-bend to 26 ft or tighten the bend to bring it under 500 ft.
- **Gate.** Appendix D requires:
 - at least 20 ft of width for a single gate (24 ft is provided);
 - a UL 325 operator;
 - ASTM F2200 construction;
 - approved emergency operation.
- **Load.** Appendix D guidance assumes 75,000-lb fire apparatus. The pavement is designed for 80,000 lb. Tunnel roofs under any fire lane are designed for the same load.
- **△ Hose reach.** The SE and SW corners of the walkout may be more than 150 ft of hose from the court.
 - **Fix [R]:** request the extension the fire code allows for sprinklered buildings, since the house has NFPA 13D sprinklers with extended coverage.
 - **Fallback:** a 20-ft reinforced-turf emergency lane on the east shoulder, kept clear of the septic field.
- **Porte-cochère.** Its 12-ft width is fine because the drop-off loop is not designated as fire access. The 24-ft court ring is.

3.6 Fire water: the new catch

Without hydrants, the fire official may size the fire reserve by building volume.

- **My rough estimate [A]:** 13,000–32,000 gal. That comes from the NFPA 1142 method: building volume $\div$ 7 (the occupancy factor for a dwelling) $\times$ a construction factor of 0.5–1.0. The low end is the 184,000-cu-ft heated box at 0.5; the high end adds the ~40,000-cu-ft attic at 1.0.
- **What it means:** the reserve could take most or all of a 30,000-gal cistern, leaving little for toilets, laundry, irrigation, and the brook.
- **Two things can push it higher:**
 - wood framing above the walkout, which can raise the construction factor to 1.5 (about 39,000–48,000 gal);
 - an exposure multiplier ($\times 1.5$), if the official counts another building within 50 ft.

Keeping ICF on all three floors helps here too.

- **Actions [F]:**

1. Get the official's number while you're checking out land.
2. Ask whether the NFPA 13D sprinklers earn a reduction.
3. Plan for about a **40,000-gal** cistern if the reserve takes most of 30,000.
4. Keep the fire reserve at the bottom of the tank: everyday pumps draw only from above the reserve line, and only the dry-hydrant line reaches the bottom.

4. Entrance Complex [F]

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ROAD =====
15' mowed strip /^^^ separator berm, ends ~35' short of connectors ^^^\
ENTRY(W)> taper 50' |<----- 215' straight ----->| taper 50' >EXIT(E)
| 14' TRAVEL LANE -----> one-way pull-through -----> |
| 14' PARK LANE [== SEMI 75' ==] STEM 24' [== SEMI 75' ==] |
|_____||_____||
wildlife break call box @cab [P1][P2] 57' [P3][P4] 20x18 ea
(W berm) ,-----' '-----, swales -> working
/ TEARDROP 96' \ stream
\ CCW, cobble /
'-----' '-----'

CARRIAGE HOUSE 26x60 [pocket]<= GATE 24' => GATEHOUSE 26x60
outer 20': porch+bench slide W; 6' partial outer 20': alcove, intercom,
inner 40': carts, tractor, cameras, pass-through lockers
plow, tools inner 40': equipment room

gate-end chamber + covered cart ramp just inside
the gate (later, with the gate tunnel; Section 13)

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Lay-by

- One way, west to east; semis pull through and never turn around.
- Two 75-ft semis park nose-to-tail, one on each side of the stem.
- Heavy-duty concrete.
- Each road connection needs county-engineer driveway approval, a sight-distance check, and flared returns wide enough for a semi.
- A second call box at cab height, private-drive signs, and cameras.

Visitor pads. Four nose-in pads outside the gate for drop-offs, solicitors, carriers, and cars left outside while their drivers ride in by golf cart.

Teardrop. Counterclockwise around a planted island of about 40 ft, ringed by a cobble apron a trailer can roll over.

Gate

- 24-ft trackless cantilever that retracts west into a wall pocket, with a 6-ft partial-open mode.
- Fire-approved emergency opener, battery backup, and manual release.
- At least 14 ft of clearance under anything spanning the drive.

Gatehouse (east; no living space)

- **Outer half:** a doorless alcove with intercom and cameras at standing and cab height. The back wall is a bank of pass-through parcel lockers: couriers load from outside, you unload from inside, and there is no door through.
- **Inner half:** equipment room (gate controller, recorder, network, fiber entry, wildlife-alert computer), with its door on the secure side only. Once the gate tunnel is built, the equipment room and the lockers' inside doors are serviced from the gate-end chamber.
- **Wired to the house** through the tunnels.
 - △ The gate tunnel comes after Phase 1, but the fiber enters here and the gate controls are needed from day one.
 - **Fix [R]:** in Phase 1, run a duct bank from the sealed gate stub under the court to the gatehouse along the future tunnel line. When the gate tunnel is built, pull the cables into its trays or leave them in the duct bank.

Carriage house (west)

- **Outer half:** a doorless porch with a bench.
- **Inner half:** carts and chargers, tractor, plow, and tools.
- Both gate buildings are shaped so no one can climb over the gate from them, and both have shielded motion lights and cameras.

Berms and swales. Planted berms curve in along the teardrop to the gate buildings. The west berm has a wildlife break on a straight stretch, lined up with a staggered gap in the tree belt. Swales along the lay-by carry runoff to the working stream.

Mailbox. Just east of the exit connector, at the spot the postmaster sets. Typical USPS guidance is 41–45 in to the bottom of the box and 6–8 in back from the road edge [A].

Wildlife alert

- A DIY thermal camera and radar feed a Raspberry Pi or Jetson running MegaDetector, which triggers LED flashers through a relay.
- **On the private drive:** use it freely, with a gate light and phone alerts.
- **On the public road:** ORC 4511.14 bars unofficial devices that imitate traffic-control devices [A].
Fix: offer the system to the road authority as a pilot feeding its own official warning sign, with flashers aimed only at drivers and off until triggered.
- **Other road-side measures:**
 - Keep habitat draws away from the road.
 - Set the tree belt back behind the mowed strip, with one natural crossing point.
 - Request official crossing signs.
 - Explore a culvert ledge or amphibian tunnels with the road authority.

5. Motor Court, Pavilion, Cistern, Walk, Porte-Cochère [F]

Court

- **Layout:** counterclockwise, 150-ft outer diameter, with a 24-ft two-lane ring that doubles as the fire turnaround.
- **Pads:** seven covered pads on 45° spokes, with the south slot left open for the axis walk. Each has wheel stops and bollard-strength posts. The pads' inner posts stand on the tunnel loop's outer wall.
- **Loop exits:** two exit stairs rise from the tunnel loop between pads on the east and west sides.
- **Walks:** an 8-ft cart-rated walk ring, with raised brick or stone crosswalks.
- **Paving:** exposed local-limestone-aggregate concrete (ODOT freeze-thaw-approved aggregate), domed about 1.5%. **Never use chloride deicers.**

Pads. Hipped S1 slate at 8:12 or steeper on white-oak frames with stone or steel post bases. **Build 4 in Phase 1 [F]**, with footings, drains, and conduit for all 7. Their power, including any future EV chargers, comes from the tunnel loop.

Pavilion

- **Size and roof:** up to about 50 ft across, with an S1 Vermont unfading green slate cone at 9:12 or steeper, stainless flashing and nails, and stainless snow rails.
- **Frame:** eight site-grown, grader-certified white-oak posts on stone bases, aligned with the spokes, and a laminated-glass lantern at the peak. Site oak goes to this frame first (\$7).
- **Finial:** if the lantern carries one, it is bronze-colored stainless steel, not copper (\$12).
- **Fireplace:** stone, on the north side. A thickened section of the tunnel loop carries the stone chimney down to firm ground, outside the cistern, and the ash pit empties out from the loop. Stainless flashing and a cricket.
- **Fit-out:**
 - Outdoor kitchen on conduit run in advance; its drain joins the house's pumped sewer line in the loop.
 - Limestone floor.
 - Retractable screens on the west side.
 - Warm dimmable lights.
- **Lightning protection:** aluminum above the roof; copper only where concealed.
- **Drainage:** a stone-lined drip ring piped to the cistern.

Cistern [F: engineered as the pavilion foundation]

- **Capacity:** 30,000 gal, with a reserved fire volume at the bottom. Plan for about 40,000 gal if the fire official's reserve takes most of 30,000 (\$3.6).
- **Shape:** as the pavilion's foundation it has to reach under all eight posts, so it is roughly **30-40 ft across and shallow**. The earlier 21-ft-wide, 10-ft-deep example was wrong for a foundation tank. Water depths (before freeboard and slabs):

Capacity	30 ft across	35 ft across	40 ft across
30,000 gal (4,010 cu ft)	5.7 ft	4.2 ft	3.2 ft
40,000 gal (5,350 cu ft)	7.6 ft	5.6 ft	4.3 ft

- **Structure:** a thickened roof slab with pier caps under the posts. The tunnel loop wraps tightly around the tank, with a flexible joint between the two structures.

- **Pipes and valves:** every tank pipe passes into the loop, and its valve sits on the loop side. That covers the inlets, outlet, overflow, pump suction, fire-hydrant line, and water-level sensor.
- **Uses:** fire reserve; toilets and laundry; drip irrigation; the brook top-up; the purified line to the NiFe vault. It overflows to the clean outfall.
- **Fire connection:** a fire-approved dry hydrant at the court lane, designed to NFPA 1142 [A: confirm the edition]. Its line leaves the tank through the loop.
- **Access:**
 - Practical entry is a side hatch from the loop into the tank, under confined-space procedures.
 - The ceremonial stone compass-rose hatch on top, aimed at the front door, stays as an inspection hatch. Its inlay is bronze-colored stainless rather than bronze, because it sits in the court's runoff path (§12).
- **⚠ Settlement risk. Fix:**
 - use a structural tank with waterstops;
 - keep the flexible joint between the tank and the loop, and give every pipe a flexible connection;
 - let the loop, not the tank, carry the chimney and the pads' inner posts.

Switch to an offset tank if the geotech finds uneven bearing.

Covered walk. Gabled slate at 8:12 or steeper, 12 ft wide and 14 ft clear, with an 8-ft path. It has a lit ceiling, and its posts are strong enough to stop a car: bollard-rated steel cores inside the stone or oak cladding [F].

Porte-cochère. The one-way drop-off loop passes under a slate porte-cochère at the north door, 12 ft wide and 14 ft clear. It has a lit ceiling and car-stopping posts like the walk. The house cart bay (§13) sits below it.

6. Garage, West Loop, Battery Basement

Garage [F]

- Extra-large two-car pull-through, about 48 ft east-west by 30 ft. Its drive-through runs along the contour, with big doors on the east and west.
- **Floor level with the house main floor,** so the walk to the house has no steps.
- **Walk-in doors on the north and south sides,** along a 6-ft cart aisle kept clear across the garage. The south door opens onto the covered walk to the west porch.
- A spur off the west side of the court makes its own loop through the garage and back. A **covered ramp** runs from the garage loop down to the garage lobby, the tunnel chamber below (§13).
- **Roof:** slate on the north slope. The south slope is dark gray standing seam carrying about 10–12 kW of PV, DC-coupled to the LFP bank.
- EV chargers, and a covered slate walk to the west porch with a lit ceiling and car-stopping posts.

Battery basement [F]

- **Structure:** two concrete vaults, a fire-rated wall between them, and a concrete ceiling under the garage slab.
- **Ventilation:** each vault vents to outdoors on its own. A gas detector boosts its fan: hydrogen in the NiFe vault; failure off-gas in the LFP vault, with deflagration venting as required. Keep the vents clear of the propane tank (§9).

- **Access and layout:** an exterior stair and a lift-out equipment hatch; batteries on plinths. **The vault doors never open into the tunnel.**
- **Drainage:** the NiFe vault floor drains to a neutralizing sump, and the footing drain runs by gravity to daylight.
- **Electrical:** the inverters and electrical panels sit in a room between the garage lobby and the battery vaults, serviced from the tunnel. AC feeders to the house run through the tunnels.
- **LFP vault (Phase 1):** 60–80 kWh, capped at 80, built from listed units of 20 kWh or less each.
- **NiFe vault:** the shell is built in Phase 1. The batteries go in **only after written approval:** 60–80 kWh nominal, 48-V strings, programmable inverters, shunt monitor, automatic watering with a purified line from the cistern, and an electrolyte change every 7–10 years.
- **Insurance [F]:** get quotes on the battery vaults before committing to nickel-iron (\$18).

Ohio code position (verified, September 2026)

Item	Status	What it means for you
Residential code	The 2019 RCO (2018 IRC basis; amended Mar. 1, 2024) governs. The Board of Building Standards' five-year review proposes "amended, new and no change rules to the 2019 Residential Code of Ohio." It went to the Common Sense Initiative office Aug. 21, 2026; comments close Sept. 30, 2026; the hearing date is still "coming soon." A Sept. 23 revision would remove the IRC electrical chapters.	No jump to the 2021 or 2024 IRC battery rules is imminent; track the final rule
RCO R327	Batteries must be "listed and labeled for residential use in accordance with UL 9540" and may not be in habitable space	LFP complies; NiFe does not as written
NEC 480	2023 NEC 480.3: "Storage batteries and battery management equipment shall be listed. This requirement shall not apply to lead-acid batteries." The 2026 NEC 480.2 also exempts vented lead-acid and nickel-cadmium, but not nickel-iron. Ohio's residential use of the 2023 NEC dates from 2024 [A]	NiFe needs a listing or an approved alternative
Future risk	2021 IRC R328 limits units to 20 kWh each and 80 kWh total in a detached garage, with 3-ft spacing	⚠ The combined 160 kWh would exceed that total if Ohio adopts it
Fire code	IFC 1207 thresholds: nickel-cadmium/nickel-metal-hydride 70 kWh; "other battery technologies" 10 kWh	NiFe may trigger fire-code energy-storage rules at 10 kWh

NiFe approval path

1. In Phase 1, present the vault design to the building official, the electrical plans examiner, and the fire official.
2. **Preferred route:** an NRTL field evaluation of the assembled bank, which produces a field label.
3. **Alternative route:** approval as an alternative material or method [A], supported by an engineer's report and NFPA 855 principles.
4. Get the approval in writing, stating the maximum kWh.
5. Get insurance quotes on the vaults with NiFe installed. If the premium or exclusions are unacceptable, stop here.
6. If an 80-kWh total cap is ever adopted, document the vaults as separately fire-rated rooms or split the capacity (for example, 40 kWh LFP + 40 kWh NiFe).
7. If the answer is no, use the vault for a future listed chemistry or as an expansion room.

7. House

Program [F]

- 120 × 48 ft, three levels, floor-to-floor heights of 11/11/10 ft.
- About 17,280 sq ft gross; about 14,900 sq ft daylit.
- **Width [F]:** 120 ft until the energy model and a professional estimate come back. Then decide 100, 110, or 120 ft. Each 10 ft changes the area by 1,440 sq ft.
- 7 suites, 8 full baths, 2 powder rooms, an elevator, and 2 stairs.
- Aging in place: 36-in doors, zero-step entries, curbless showers. The garage floor is level with the main floor, and the house cart bay is level with the elevator lobby.

Room schedule and plans [F: these govern]

My earlier sketches drifted from the original room layout. The stair and elevator no longer lined up floor to floor, and the library lost its south-facing spot. The plans below put the original rooms back on one 120-ft grid, drawn at 1 character = 1 ft east-west (numbers are feet from the west wall). Depths are not to scale. The rules they follow:

1. **One core, stacked.** A 24-ft core bay at 48–72 ft holds the same three pieces on every level:
 - the limited-use elevator on the west (48–56);
 - an 8-ft hall on the axis (56–64);
 - the main stair on the east (64–72).

The cupola sits over it on the axis.

2. **Service stair stacked.** It runs at 88–96 ft between the main and second floors, beside the scullery and the laundry. The walkout is reached by the main stair, the elevator, the tunnel, and its own south doors.
3. **Same bands on every level.** A 20-ft north band and a 28-ft south band, matching the buried service strip and daylit band of the walkout. The buttress crosswalls fall on room lines.
4. **Library faces south.**

Sizes are gross, measured to wall centerlines, and include a share of the east-west hall that runs between the bands.

Walkout level: 28-ft daylit south band + 20-ft buried service strip

0	24		48		72		96		120	
+	+		+		+		+		+	
	THEATER 24x20		WINE 10x20		PARTS LIB 14x20		ELEV LU/LA		LOBBY N->bay	
					STAIR main		SAFE RM 12x20		ELEC/IT 10x20	
									MECHANICAL 26x20 heat tank	
+	+		+		+		+		+	
	GUEST SUITE 24x28		SUITE 5/FLEX 24x28		GALLERY 24x28 (axis)		REC ROOM + BAR 30x28		GYM+BATH 18x28	
+=====+										
covered patio under 6' dry-under deck; ONE sleeved geo header (S wall)										
LOBBY opens north to the HOUSE CART BAY (tunnel), level with the elevator										

- **Parts library** (14 × 20 ft) sits where cold storage was; the wine cellar stays beside it.
- **Mechanical room** (26 × 20 ft) now holds the enlarged heat storage tank. That works because the drinking-water and sprinkler tanks moved outside, buried beside the well (\$9).
- **Safe room** is built to FEMA P-361 / ICC 500, next to the core.
- **Electrical/IT** sits between the safe room and mechanical, a short run from where tunnel power enters at the cart bay.

- The **gallery** carries the axis from the core to the covered patio. The gym holds the eighth full bath.
- The in-house battery room is **deleted**.

Main level

0	24	48	72	96	120					
+-----+-----+-----+-----+-----+-----+										
	PRIMARY SUITE	MUDROOM	PWDR x2	ELEV	FOYER	STAIR	BUTLER PANTRY	SVC	SCULLERY/PANTRY	
	(west wing)	12x20	10x20	LU/LA	entry	main	16x20	STAIR	24x20	
+-----+-----+-----+-----+-----+-----+										
	26x48	LIBRARY/	GREAT ROOM		DINING		KITCHEN/BKFST			
		OFFICE	32x28 (axis)		18x28		26x28			
		18x28 S								
+=====+=====+=====+=====+=====+=====+										
north: entry on axis under porch/porte-cochere; 10' wrap porch at grade										
south: 6' dry-under deck (membrane + edge gutter); 20x20 corner decks										

The arrival sequence hides the valley until you reach the great room. The front door is on the axis, and the foyer runs straight south into the great room, with the elevator and main stair to either side. The **library** is back on the south side, between the primary suite and the great room. The butler's pantry serves the dining and great rooms from the scullery side. The mudroom opens to the north wrap porch, which wraps to the west porch and the covered walk from the garage.

Second floor

0	24	48	72	96	120					
+-----+-----+-----+-----+-----+										
	S1 BATH/CLOSET 24x20		S2 BATH/CLOSET 24x20		ELEV LANDING STAIR LU/LA cupola main		S3 BATH/CL 16x20		SVC LAUNDRY S4 BATH/CL STAIR 10x20 14x20	
+-----+-----+-----+-----+-----+										
	SUITE 1 24x28		SUITE 2 24x28		LOUNGE/LOFT 24x28 (axis)		SUITE 3 24x28		SUITE 4 24x28	
+-----+-----+-----+-----+-----+										
=====										
cupola over the core landing (rain-sensing vents)										

Other house details [F]

- The porch ends connect to the berm-top path.
- A cupola over the stair core, on the axis, with rain-sensing vents that close automatically. Its finial is bronze-colored stainless steel, not copper.
- Finished grade along the south wall within 12 in of the walkout slab, to satisfy the story-count rule.
- The arch stays unlocked as the emergency exit route: at least 44 in clear, and lit.

Design loads [F]

- Ultimate design wind speed 115 mph. Open roofs are designed to Risk Category III (\$10).
- Ground snow load about 20 psf.
- Footings at least 36 in deep (frost depth).
- EPA Radon Zone 1, the highest-risk zone.

Structure and envelope [F]

- **Walls:** ICF on all three floors if the budget allows. The fallback is ICF for the walkout's retaining "bathtub" with wood framing above it. Decide at the first professional estimate. Note that wood framing above the walkout can raise the fire-water reserve (§3.6).
- **Floors and walls:** a precast hollow-core plank first floor, which also braces the building, plus concrete buttress crosswalls in the service strip.
- **Footings:** to native soil or rock per the geotech, and at least 36 in deep.

- **Waterproofing and drainage:** a fully adhered membrane with drainage board, gravity footing drains, and an uphill curtain drain.
- **Concrete:** slag-blend where the engineer allows.
- **Radon and air quality:**
 - Active sub-slab radon system and continuous monitors.
 - ERVs balanced slightly positive on the walkout level.
 - Sealed, gasketed grinder basin.
 - Sealed-combustion masonry heater with outdoor air, the only combustion appliance in the house.
 - Induction cooking; no gas enters the house.
 - Interlocked kitchen make-up air.
- **Tunnel connection [R]:**
 - The door between the walkout lobby and the house cart bay is treated as an exterior door: gasketed, self-closing, and fire-rated [A].
 - The cart bay stays outside the air barrier.
 - Radon and CO monitors sit on the bay side.
- **Testing and targets:** blower-door and worst-case depressurization tests; about 1.5 ACH50; ducts inside the envelope.

Sun angles at 39.4°N (my calculation)

Noon sun altitude is 74.0° at the summer solstice, 50.6° at the equinox, and 27.2° at the winter solstice. At those angles, the 6-ft deck over the walkout glass shades:

Date	Shade reaches down	Effect
Summer solstice	about 20.9 ft	Glass fully shaded
Equinox	about 7.3 ft	Partly shaded
Winter solstice	about 3.1 ft	Only the top ~1.6 ft of glass shaded

For the upper glass, the overhang depth is roughly the glass height divided by $\tan 74^\circ$. That works out to about 2.6 ft for 8-ft glass with the overhang 1 ft above the head.

Roofs [F]

- Main gable at 7:12 with the ridge running east-west.
- **North slope:** S1 slate.
- **South slope:** dark gray standing seam with clamp-on PV. The energy model sets the size; 40-50 kW is the default [R]. About 50 kW is the practical maximum by geometry (120 × 31 ft, ~70% usable, ~20 W/sq ft).
- **PV-roof safety:** a snow rail at the PV eave, fall-protection anchors, and service lanes.
- **Gutters:** stainless 6-in half-round with 4-in leaders and overflow notches on the outer face.
- **Snow rails:** engineered stainless rails over every walk, door, deck, and parking edge.
- **Protection:** aluminum lightning protection (copper only where concealed) and whole-house surge protection.

Windows [F]

- **Frames:** aluminum-clad wood.

- **Glass:** triple glazing, with high solar gain on the south and low gain on the east and west.
- **Replaceable units:** glass units sit in removable interior stops, so a unit can be swapped when its seal fails without replacing the sash. Standard unit sizes where the design allows; every size and maker goes in the house book.
- **Bird-safe glass:** a frit or etched pattern at no more than 2-in spacing on large panes.

Exterior [F]

- **Stone:** local limestone on the lower walls, piers, arch, and gate buildings.
- **Siding:** white-oak heartwood over a rainscreen gap, with deep overhangs. It is site-milled from your own oak if any is left after the pavilion frame. Otherwise it is bought heat-treated (thermally modified) wood, which also covers any shortfall.
- **Timber:** dried properly, and used structurally only if grader-certified.

Site oak allocation [F]. The legacy woods are never harvested, so site oak comes only from trees cleared for the building area; the land decides how much there is. It goes:

1. first to the pavilion frame;
2. then to the cabinets;
3. then to the siding.

Bought wood covers any shortfall: heat-treated wood for siding, and graded white oak for the pad frames and floors.

Interiors [F]

- **Walls:** plaster (lime or veneer plaster over a solid base), not painted drywall.
- **Cabinets:** built from your own white oak, in solid wood with standard hardware.
- **Floors:** polished concrete in the walkout's sunny band; white oak above.
- **Lighting plan:** by the architect or a lighting designer, room by room:
 - layered ambient, task, and accent light;
 - screw-in sockets only, no integrated LEDs;
 - warm 2700 K lamps indoors;
 - plain wall switches and dimmers, with smart controls in the electrical room.

Elevator [F]. A commercial limited-use/limited-application (LU/LA) unit.

- Its 25-ft travel limit covers the house's 22 ft (walkout to second floor).
- Its parts are standard, so any elevator company can service it.
- It stops at the walkout lobby (level with the house cart bay), the foyer, and the second-floor landing.
- Confirm the state permit and inspection path [A] (§16).

Kitchen and cooking [F]

- Induction cooking.
- The early design assumed gas; induction keeps combustion out of an airtight house that's managing radon.
- The hood's make-up air is interlocked.
- Nothing in the house burns gas. Gas or propane serves only the generator (and a pavilion grill, if wanted).

8. Fixtures, Devices, Parts Library, House Book [F]

Rules

- Commercial or institutional grade.
- Solid brass, bronze, stainless, cast iron, vitreous china, fireclay, or stone. No plated plastic. (Inside the house only; nothing copper-bearing goes in a runoff path, \$12.)
- Standard sizes and parts.
- Mechanical controls at the point of use; electronics in the electrical room.
- One model or one internal-parts family per category, house-wide.
- Everything reachable.

Electrical

- **Devices:** hospital-grade outlets and specification-grade switches.
- **Wiring:** screw terminals (no push-in) in extra-deep metal boxes, with branch wiring in EMT conduit so it can be re-pulled later.
- **Protection:** GFCI/AFCI in the breakers, not the outlets.
- **Plates:** stainless or brass.
- **Fixtures:** screw-in sockets, no integrated LEDs.
- **Controls:** smart controls in the electrical room, plain switches in the walls.
- **Alarms:** leak sensors with automatic main shutoff, plus pump, cistern-level, radon, tunnel-sump, and tunnel high-water alarms.

Plumbing

- **Toilets:** vitreous china, 12-in rough-in.
- **Tubs and sinks:** enameled cast-iron tubs; fireclay or cast-iron sinks; soapstone kitchen and scullery sinks.
- **Faucets and valves:** solid-brass, rebuildable, on a single cartridge standard.
- **Shutoffs and drains:** quarter-turn brass shutoffs, brass traps, and cleanouts at every fixture group.
- **Supply:** a PEX home-run manifold, hot-water recirculation, and a geothermal desuperheater.
- **Cistern water:** cast-iron drain stacks, and a separate, labeled non-potable line from the cistern to the toilets and laundry (\$9).

Hardware. Brass or bronze mortise locks with interchangeable cores; ball-bearing hinges.

Parts library (walkout, where cold storage was)

- **Conditions:** earth-stable temperature, about 45% humidity, steel shelves, labeled bins.
- **Stock:**
 - 10-20% spares of high-count items.
 - Complete spares of breakables (glass shades, a toilet, a sink).
 - One of each breaker type.
 - Attic stock of slate, tile, oak, limestone, pavers, plaster, and paint formulas.
- **Storage:** heat-sealed bags with desiccant; VCI bags for metal parts.
- **Rotation:** rubber parts every 10-15 years.

House book

- **Item codes:** codes such as E-SW-01 and P-CT-02 appear on bins, in the book, and on a tag at each installed item.
 - **For each code:** maker, part numbers, locations, quantity, bin, date stored, and current equivalents.
 - **Records:**
 - A surveyed as-built of every drain, bore, conduit, cleanout, valve, spreader elevation, and tunnel run, with each tunnel cable and pipe labeled.
 - Window glass-unit sizes and makers.
 - The planting map.
 - The covenant and the recorded conservation easement.
 - The maintenance plan.
 - **Format:** archival paper, PDF/A, and a spreadsheet, with an off-site copy. It conveys with the house.
-

9. Systems

HVAC [F]

- **Ground loop:** vertical bores, drilled before the berms go in and placed outside the basin and berm fill. The energy model sets the count (about 15–21 is the planning range). The bore field is south of the house, and the headers enter through **one** sleeved penetration in the exposed south wall. They never run in the tunnels.
- **Distribution:** zoned by floor and wing, with radiant slabs on the walkout level.
- **Heat storage tank:** sized by the energy model. It has grown past the early 1,000–2,000-gal buffer, which is why the mechanical room was re-planned (§7). My check: $1,000 \text{ gal} \times 8.34 \text{ lb/gal} \times 40^\circ\text{F} \approx 334,000 \text{ BTU} \approx 98 \text{ kWh}$, so about 100 kWh per 1,000 gal.
- **Ventilation:** three ERVs totaling about 580 cfm, which is consistent with ASHRAE 62.2 (≈ 586 cfm) [A].
- **Also:** dehumidification, the masonry heater, and the concrete thermal mass.
- **Optional heat loop [R]:** an insulated hot-water loop from the heat tank through the tunnels, to keep the garage basement and gatehouse above freezing.

Energy and electrical [F]

- **Utility service:** ask the utility to run its own line in its own conduit to a transformer beside the court. **Everything past the meter runs in the tunnels:**
 - the house service;
 - the garage and its inverters;
 - the gatehouse;
 - the pavilion;
 - court lighting;
 - the parking pads, including any future EV chargers.
- **Service:** sized by an NEC 220 load calculation and the energy model, with conduit ready for 800 A.
- **Panel:** a smart panel with load management; empty conduit to every floor and outbuilding.

- **Generator:** UL 2200-listed and installed per NFPA 37, sized by the energy model. It stands beside the transformer at the court [R] and runs on natural gas or propane.
- **Solar:** south slopes only, sized by the energy model. At about 1,200 kWh per kW per year [A], 62 kW of PV would produce about 74 MWh a year.
- **Energy priority:** thermal tank → EVs and timed loads → LFP → NiFe → grid. Draw on LFP first.
- **Inverters:** each battery bank on its own inverters (UL 1741), AC-coupled, never paralleled on the DC side.
- **Tax credits:** 26 USC 25D "shall not apply with respect to any expenditures made after December 31, 2025." The \$45L and \$30C credits also ended June 30, 2026. Budget with no credits.

Water supply [F]. Public water if available. Otherwise:

- a well on the west side, with setbacks from the septic, fuel, and vaults [A: OAC 3701-28];
- 1,000–2,000 gal of drinking-water storage, **buried outside beside the well**, not in the mechanical room;
- the well and tank lines enter the garage tunnel through a short sleeve and run to the house mechanical room, where treatment and pressure equipment stay;
- treatment matched to the water tests.

Cistern water [F]. The cistern supplies the toilets and laundry, drip irrigation, and the brook top-up. A purified line runs to the nickel-iron battery vault. The non-potable system gets labeled pipe, filtration and disinfection as the plumbing code and health district require, and backflow protection [A].

Wastewater [F]. Sewer if available. Otherwise septic on the east shoulder with a reserve area; the soil evaluation sets the type (mound, drip, or aerobic).

- The walkout level drains to a duplex grinder station with a battery-backed alarm.
- The house's pumped sewage line runs through the main tunnel and the loop, and leaves the loop on the east side through a short sleeve to the septic.
- The pavilion kitchen drain joins it.

Fire [F/R]

- NFPA 13D sprinklers with extended coverage in the garage, attic, porches, and mechanical spaces [R]. The fire official decides how much of the house they cover.
- A **sprinkler tank buried beside the well**, sized by the sprinkler designer. The sprinkler main runs through the tunnels to the house.
- Access lanes at least 20 ft wide (26 ft if the dead end is over 500 ft), with 14-ft clearances.
- The cistern fire reserve and dry hydrant (\$3.6, \$5).
- Lightning protection on all four structures.

Fuel [F]

- **Source:** natural gas if a main runs along the road; otherwise a 1,000-gal buried propane tank.
- **What it serves:** the generator (and a pavilion grill, if wanted). Nothing in the house burns gas.
- **Propane tank siting.** It needs re-siting now that the garage moved. Keep it:
 - at least 10 ft from any building;
 - clear of the battery-vault vents, the creek, the septic field and reserve, and the well (isolation distance per OAC 3701-28 [A]);
 - within the supplier's delivery-hose reach of the drive [A].

A candidate spot is the northeast quadrant of the hilltop, beside the S-bend and north of the septic field, reached from the drive.

- **Route:** propane stays out of the tunnels. It's heavier than air and would pool there. Its line is buried separately along a marked route to the generator.

Internet and data [F]. Fiber internet, entering at the gatehouse, with Starlink as a backup. Data, network, cameras, intercom, gate controls, and the wildlife alert run in the tunnels.

Security and technology [F]. Everything is controllable from the house and your phones. Outdoor lighting is dark-sky, 2700–3000 K, and motion-sensed, except the covered walks and porte-cochère, which have lit ceilings.

10. Integrated Water Management

Three streams [F]

1. Roof water → cistern.

- Passes through leaf screens and first-flush diverters.
- Stainless leaders run on standoffs or inside hollow stone piers, **never inside timber posts**.
- The cistern supplies toilets and laundry, gravity-feeds drip irrigation, tops up the brook reservoir, and supplies the NiFe watering system through the purified line. It overflows to the clean outfall; the overflow pipe leaves through the tunnel loop.

2. Pavement water → planted bioretention creeks.

- Oil-absorbent sump inserts at the drains.
- Trench drains at the court/forecourt seams and along the porch edge.
- Lay-by and pad runoff goes to swales.
- Everything ends at the **working outfall**: a 40–60-ft spreader into a filter strip, at least 200 ft from the pool where possible.

3. Hillside water → routed around everything.

- A stone-lined interceptor creek.
- Step-pool side creeks, with check dams spaced at drop ÷ slope (about 12.5 ft at 12% and 19 ft at 8% for a 1.5-ft drop).
- Every path ends at the bottom of the property.

Storm drains and creek pipes stay out of the tunnels. They need steady slopes and would flood a tunnel if they clogged, so they keep cleanouts at the surface (§13).

Berm Ring storage check (planning level)

- **Geometry [F]:**

- About 7 ft tall, with a 10-ft crest.
- 3:1 inner slope and 2:1 outer slope.
- Keyed compacted fill with a clay core and a toe buttress.
- Yard about 140 × 70 ft draining to a 140 × 25-ft basin.

- **Design depth:** NOAA Atlas 14 Volume 2, version 3, partial-duration series, at 39.43°N, 84.21°W:

Storm	24-hr depth	90% confidence interval
10-yr	3.96 in	3.70–4.23 in
25-yr	4.63 in	4.32–4.94 in
100-yr	5.70 in	5.29–6.09 in

The larger of 6.09 in and 5.70×1.2 governs, so the design depth is **6.84 in [R]**.

- **Volume needed:** about 19,300 sq ft drains into the ring. At 100% runoff with the outlet blocked, that is $19,300 \times 6.84 \div 12 \approx \mathbf{11,000\ cu\ ft}$.
- **Volume available:** the basin holds about 8,750 cu ft and the lower-yard wedge about 2,500 cu ft, for roughly 11,250 cu ft total. That barely meets the need.
 - $\triangle$ **Fix [R]:** make the basin average at least 2.75 ft deep below the storage line, or widen it to 30 ft. The civil engineer re-runs the check with the surveyed dimensions.
- **Outlet:** an unlocked walk-through stone arch (also the emergency exit route) plus an armored spillway. Do a burrow watch every year.
- **Dam safety:** ORC 1521.06 exempts any dam "regardless of height, that has a storage capacity of not more than fifteen acre-feet." The ring holds about 0.25 acre-ft. Get written confirmation from ODNR Dam Safety.
- **Atlas 15 [A]:**
 - NOAA scheduled preliminary estimates for the lower 48 states for peer review by September 2026, with publication in 2027.
 - I found no confirmation that they had been released yet.
 - Re-check the design if the published site depth exceeds 6.84 in.
 - The Warren County Engineer's manual cites the older Bulletin 71 rainfall depths. Use whichever source is more conservative.

Zen brook [F]

- Downspouts pour into carved stone basins, which feed a shallow, 6–12-in river-stone brook winding to the basin.
- It reads as a dry streambed between rains.
- **Trickle reservoir [F: include]:** a reservoir at the foot, topped up from the cistern through the tunnel's cistern line, with a solar trickle pump.
- Foundation drains stay piped to the arch, not into the brook.

Clean outfall [F]. The Berm Ring outlet and cistern overflow go to a 60–100-ft level spreader. From there, water reaches the seasonal pool, which has a low-flow bypass so it dries by mid or late summer, and then spreads as sheet flow across the meadow. **No fish in the seasonal pool, ever [F]:** it stays fishless so amphibians can breed. Build the spreaders and pool **last**.

No-chloride covenant [F]. Chloride deicers are banned permanently. Use sand or grit; potassium or calcium-magnesium acetate is allowed only on short ramps during ice storms. Record the ban in the house book and in the conservation easement (§14).

Construction permit (verified)

- Ohio EPA CGP OHC000006 took effect Apr. 23, 2023 and **expires Apr. 22, 2028**.
- The SWP3 (stormwater pollution prevention plan) must document the ring, creeks, and spreaders as permanent post-construction controls. The permit's water-quality volume is based on "a 0.90-

inch rainfall."

- **△ Fix:** put the renewal on the calendar.

Storm-proofing [F]

- Every drain has an above-ground overflow path.
 - Open roofs are designed to Risk Category III wind.
 - No downspouts onto walks; pipes below frost depth (36 in) or self-draining.
 - Both cart ramps are roofed, with drains at the top and bottom, so rain can't run into the tunnels.
-

11. Landscape and Ecology [F]

Zen garden

- **Upper yard:** raked gravel, fossil-limestone specimen stones from the excavation, and benches.
- **Middle yard:** the brook and stepping stones.
- **Basin:** sedges, blue flag iris, cardinal flower, swamp milkweed, buttonbush.
- **Small trees, at the sides only:** hornbeam, serviceberry, redbud, pagoda dogwood, witch hazel, fringe tree. None go on the berm faces.
- **Ground layer:**
 - Sedges and prairie dropseed in sun.
 - Moss in shade.
 - Restrained flowers.
- **Paths:** a level crushed-limestone loop that works for carts and wheelchairs.
- **Benches:** stone slab (it won't float in a flood) or white oak.
- **Lighting:** low shielded path lights.

Berm plantings

- **Outer faces:** shrubs and small trees only: redbud, serviceberry, arrowwood viburnum, ninebark, 'Taylor' red cedar.
- **Crests and inner faces:** nothing woody.
- **Tall evergreens:** planted beyond the outer toes. Red cedar and white pine are favored; 'Green Giant' only where a fast screen is essential; Norway spruce only beyond the toe.
- **Arch and spillway:** grass and stone only.

Privacy belts. Along the road, both side lines, and the south line, with shorter red cedar in the view corridor so the view stays open.

Legacy woods (planted first)

- **Placement:** at least 100 ft from buildings.
- **Oak backbone:** white, bur, chinkapin, and swamp white oak.
- **Companions:** shagbark hickory, sugar maple, black gum, tulip poplar, and buckeye.
- **Planting:** bare-root seedlings and local acorns, with tree tubes and early grass control.
- **Meadow oaks:** 3-4 open-grown bur or white oaks at least 60 ft apart.
- **Protection:** the conservation easement (\$14), and no timber harvest.

Meadow and wildlife

- A native meadow over the view corridor and septic field, mowed once in late winter.
- Snags, logs, and brush piles; bluebird boxes; bat houses on the garage.
- Native shrub corridors along the creeks.
- Pawpaw, persimmon, serviceberry, elderberry, and hazelnut along woodland edges, with an option to take part in the American Chestnut Foundation program.
- A fishless seasonal pool (\$10).
- Invasive removal first: bush honeysuckle, callery pear, garlic mustard, stiltgrass, wintercreeper, tree-of-heaven.

Trails. An 8-ft crushed-limestone cart loop, plus the berm-top path with a gentle ramp. Carts fit through the arch.

12. Materials Schedule [F]

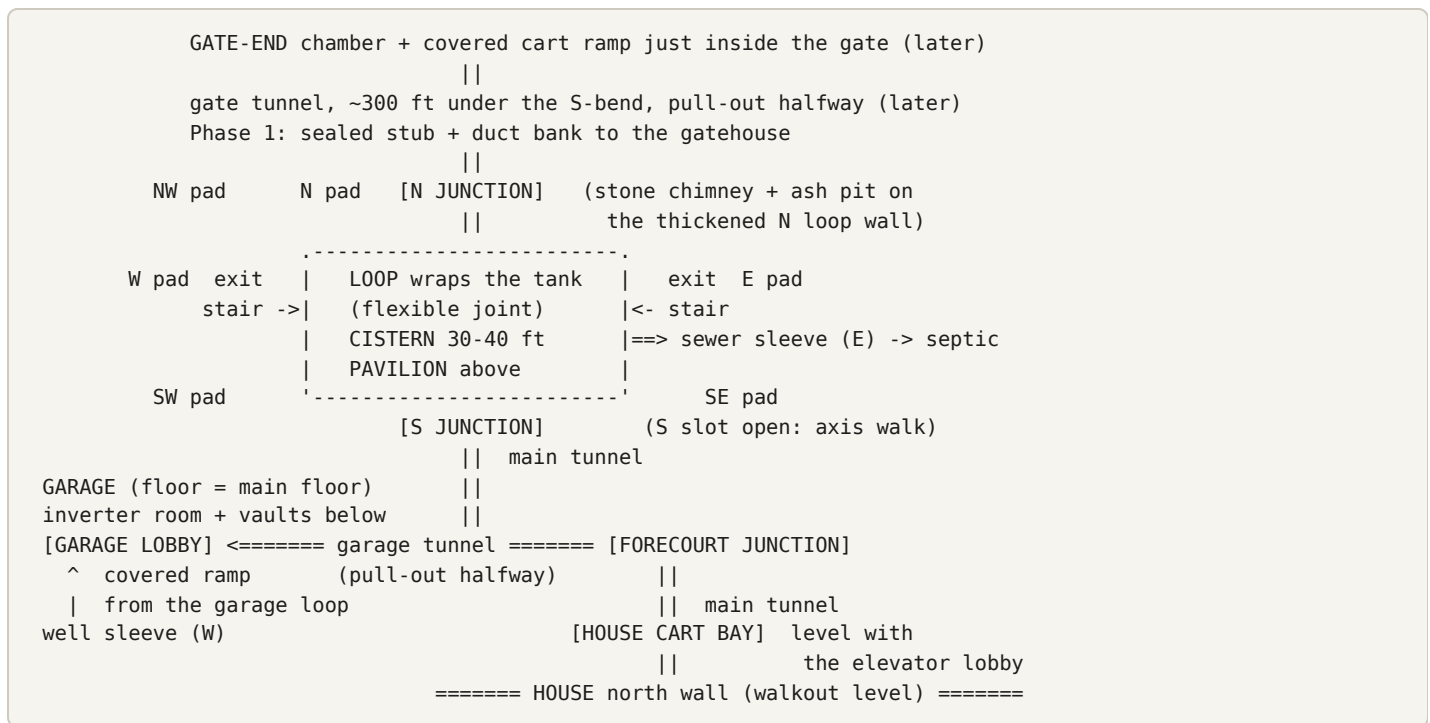
Material	Specification	Where used
Slate	S1 grade (ASTM C406), Vermont unfading green; stainless nails and flashing; snow rails	Pavilion (≥9:12), pads, covered walk, porte-cochère (≥8:12), gate buildings, north slopes
Standing seam	Dark gray, clamp-on PV	South slopes only
Gutters and leaders	Stainless	Everywhere; no copper in any runoff path
Finials and exterior inlays	Bronze-colored stainless steel (replaces copper and bronze)	Cupola finial; pavilion lantern finial, if used; compass-rose hatch inlay
Lightning protection	Aluminum exposed; copper only concealed	All roofs
Limestone	Local, from the excavation and quarried	Walls, piers, arch, floors, check dams
White oak	Site-grown from the building-area clearing, grader-certified; bought graded oak covers shortfalls	Pavilion frame first, then cabinets, then siding; pad frames and floors
Thermally modified wood	Bought, heat-treated	Siding shortfall
Windows	Aluminum-clad wood; triple glass units in removable stops	House, gate buildings
Plaster	Lime or veneer plaster	Interior walls
ICF + precast plank	Engineered	House (all three floors if the budget allows)
Concrete	Slag-blend; exposed limestone aggregate	Structure, paving, tunnels
Excluded	Copper roofing, gutters, and finials; exposed copper or bronze in any runoff path; gas appliances in the house; polycarbonate; fiber cement	—

13. Cart and Utility Tunnels [F]

Below grade, a loop around the cistern connects to tunnels running to the house, the garage, and (later) the gate. Carts use them in any weather, and every utility past the meter runs in them where it

can be reached for service.

13.1 Hilltop layout (schematic; north at top; not to scale)



13.2 The loop around the cistern

- **Connections:** the loop meets a chamber on the south side (main tunnel) and one on the north side (gate tunnel). The loop itself works as a turnaround, since carts can just drive around it.
- **Tank pipes:** the loop wraps tightly around the tank, with a flexible joint between the two structures. Every tank pipe passes into the loop, and its valve sits on the loop side:
 - inlets;
 - outlet;
 - overflow;
 - pump suction;
 - fire-hydrant line;
 - water-level sensor.

A side hatch gives entry into the tank itself.

- **Chimney:** the loop section under the pavilion chimney is built thick enough to carry the stone chimney down to firm ground. The fireplace's ash pit also empties out from the loop.
- **Posts and exits:** the parking pads' inner posts stand on the loop's outer wall. Two exit stairs rise between pads on the east and west sides.

13.3 Turnaround spots

- **Chambers:** about 24 × 24 ft each, at six spots:
 - the house cart bay;
 - the forecourt junction;

- both loop junctions;
- the garage lobby;
- the gate end.

A golf cart can U-turn in one; a utility vehicle needs a three-point turn.

- **House cart bay:** it sits level with the elevator lobby, so there are no steps between cart and elevator.
- **Pull-outs:** side bays halfway along the garage and gate tunnels, for passing and three-point turns.

13.4 Cart entrances

- **Garage now:** a covered ramp from the garage loop down to the garage lobby.
- **Gate later:** a second ramp just inside the gate, built with the gate tunnel.
- **Size:** on flat ground, each ramp runs about 90–110 ft at a gentle slope. Where space is tight, a platform lift is the fallback.
- **Weather:** both ramps get roofs and drains at the top and bottom so rain can't run in.

13.5 Utilities in the tunnels

Water lines run low on one wall; power and data run high on the other.

- **Power:**
 - Everything past the meter runs in the tunnels: the house service, the garage and its inverters, the gatehouse, the pavilion, court lighting, and the parking pads, including any future EV chargers.
 - Ask the utility to run its own line in its own conduit to a transformer beside the court.
- **Data:** fiber, network, cameras, intercom, gate controls, and the wildlife alert.
- **Water:**
 - Well water enters the garage tunnel through a short sleeve. So do the lines from the buried drinking-water and sprinkler tanks beside the well.
 - Cistern water supplies the toilets and laundry, irrigation, and the brook top-up.
 - A purified line runs to the nickel-iron battery vault.
- **Fire:** the sprinkler main and the fire-hydrant pipe.
- **Sewer:** the house's pumped sewage line to the septic, which leaves the loop on the east side through a short sleeve. The pavilion kitchen drain joins it.
- **Heat (optional):** an insulated hot-water loop from the house's heat tank, to keep the garage basement and gatehouse above freezing.

13.6 Serviced from the tunnels

- The cistern valves and hatch.
- The chimney ash pit.
- The garage's inverters and electrical panels, in a room between the lobby and the battery vaults. The vault doors still never open into the tunnel.
- The gatehouse equipment room and the inside doors of the parcel lockers (once the gate tunnel is built).

13.7 Stays out, and why

- **Propane:** it's heavier than air and would pool in a tunnel. It stays buried separately along a marked route.
- **Storm drains and creek pipes:** they need steady slopes and would flood the tunnel if they clogged. They keep cleanouts at the surface.
- **Geothermal pipes:** the bore field is south of the house, and its pipes come in through the south wall as planned.

13.8 Tunnel basics [R]

These fill in what the tunnel decisions above left open. The structural and civil engineers confirm them.

- **Section:** precast concrete box sections, about 10 ft wide × 9 ft clear. That's one cart lane with room to walk beside it, plus pipes and trays on the walls. Roofs under the court ring and any fire lane carry fire-apparatus loads (\$3.5).
- **Waterproofing and drainage:** the same fully adhered membrane and drainage board as the house. Floors slope to sumps at the low points, each with duplex pumps and a high-water alarm to the house panel.
- **Frost:** at least 36 in of cover where the grade allows; insulate the roof near the ramps and exit stairs.
- **Air:** electric carts only, so there's no combustion in the tunnels. Mechanical ventilation, and CO and radon monitors at the house cart bay.
- **Doors:** gasketed, self-closing, fire-rated doors [A] where the tunnels meet the house and the garage lobby. The tunnels stay outside the house's air barrier.
- **Light and exits:** screw-in ceiling fixtures on the house lighting rules, battery-backed emergency lights, and lit exit signs at the stairs and ramps. The two loop stairs, the ramps, the house, and the garage lobby give several ways out.
- **Approvals:** ask the building and fire officials how they classify the tunnels, and whether the house connection needs any separation beyond the rated door [A].

13.9 Cost and phasing (my rough estimate)

The loop has to be built at the same time as the cistern, so the first phase now includes:

- the house cart bay and main tunnel;
- the forecourt junction;
- the garage tunnel and its ramp;
- the full loop;
- a sealed stub under the court pointing toward the gate, with the Phase 1 duct bank to the gatehouse (\$4).

That comes to about **\$2.2-5.2M**, more than the earlier core estimate because the loop, ramp, and chambers all land in that first phase. The gate tunnel, its chamber, and the second ramp come later for about **\$1-2.1M**, so the total is roughly **\$3.2-7.3M**.

14. Conservation Easement [F]

- **Which acres:** the woods, meadow, pool, and outfalls. On the illustrative parcel that is:

Area	Acres
Legacy woods	≈10.8
Meadow and view corridor	≈4.0
Seasonal pool and buffer	≈0.8
Creeks, spreaders, filter strip (the outfalls)	≈0.8
Total	≈16.4 (about 60% of the parcel)

- **What stays out:** the building area. That covers the entrance complex, drive, court, tunnels and ramps, garage, house, Berm Ring, well area, lawns, and the **septic field and reserve** (which must stay repairable). The privacy belts can be added if the land trust wants them.
- **When:** record it **after construction**, so the building area can be carved out exactly from the as-built survey.
 1. At purchase, sign a letter of intent with the land trust so it can review the plans.
 2. After Phase 1 construction, have the as-built survey done and the trust's baseline documentation report prepared.
 3. Record the easement.
- **Which land trust (default; confirm at purchase):**
 - **Little Miami Conservancy**, if the parcel drains to the Little Miami (the Todd Fork and Caesar Creek valleys are in its watershed). It serves Warren, Clinton, Greene, and neighboring counties. In 2024 it placed a 25-acre easement along First Creek, a Todd Fork tributary in Warren County.
 - **Fallbacks:** Tecumseh Land Trust (accredited; Greene, Clark, Champaign, and Madison counties) or Three Valley Conservation Trust (accredited; Warren, Butler, Hamilton, Montgomery, and Preble counties).
 - For a Brown, Highland, or Adams County parcel, find the local trust through the Land Trust Alliance directory [A].
 - I could not confirm the Little Miami Conservancy's accreditation status. Ask each trust about accreditation, its stewardship-endowment gift, and whether it will hold an easement with a carved-out building area.
- **Reserved rights:**
 - maintain the spreaders, pool, creeks, check dams, and trails;
 - remove invasives and hazard trees (in bat-safe months);
 - mow the meadow in late winter;
 - keep and repair utility lines already built.
- **Restrictions:** no timber harvest, no subdivision, no new buildings, and the no-chloride covenant (§10).
- **Taxes:**
 - **CAUV:** CAUV counts woodland only when it is devoted to commercial timber production, so the eased woods will likely leave CAUV. Land in qualifying conservation use can stay in at the lowest soil value [A].

- **Fix:** before closing, ask the county auditor which eased acres can stay in CAUV.
- Have the seller pay or credit recoupment on every acre that will leave it (\$2).
- **Federal:** a donated easement may qualify for an income-tax deduction under IRC §170(h), with a qualified appraisal [A: tax advisor].

15. Phasing Plan

Phase 1: enduring bones

1. Plant the legacy trees first; clear trees October–March; remove invasives.
2. Geotech, survey, SWP3, and erosion controls.
3. Interceptor creek, curtain drain, and side creeks.
4. Geothermal bores, before the berms go in (April–October).
5. The entrance complex, with the duct bank from the court to the gatehouse.
6. **Cistern and tunnel core together:**
 - the cistern;
 - the full loop with both junctions;
 - the sealed gate stub;
 - the main tunnel;
 - the forecourt junction;
 - the house cart bay;
 - the garage tunnel, garage lobby, and covered ramp.
7. Drive, court, and pavilion on the cistern; **4 pads**, with footings, drains, and conduit for all 7.
8. Garage with both vault shells and the inverter room; install the LFP bank.
9. House, roofs, staged PV, Berm Ring, and arch.
10. Septic, well, the buried drinking-water and sprinkler tanks, fuel (gas service or propane tank), transformer, and generator.
11. Both spreaders, last in the phase.
12. As-built survey, then record the conservation easement (\$14).

Phase 2.

- The remaining 3 pads, the pavilion kitchen, and suite fit-out.
- Pool establishment, the Zen garden with the brook's trickle reservoir, the meadow, and trails.
- The gate tunnel, gate-end chamber, and second ramp (about \$1–2.1M), in Phase 2 or later.

Phase 3. The NiFe go/no-go decision, after written approval, acceptable insurance quotes, and 1–2 years of metered data.

16. Permits and Approvals Checklist

Approval	Authority
Zoning, including front-yard accessory structures	Township or county
Driveway permits for both connections	County engineer / township / ODOT

Approval	Authority
CGP notice of intent + SWP3 (renew by 2028)	Ohio EPA; county soil and water district
Septic permit	Local health district
Well permit [A]	Health district
Building permit (RCO), including the tunnels and their house connection	Certified building department
Electrical, PV, and battery systems (NiFe: written approval or NRTL label)	Same
Non-potable cistern use for toilets and laundry [A]	Building department (plumbing) and health district
Elevator installation permit and inspection, if required for a residence [A]	Ohio Department of Commerce, elevator section
Gate opener, dry hydrant, fire-water reserve, access width, hose-reach extension, sprinkler scope, vault review, tunnels	Fire department
Propane tank (if no gas main)	Fire official; propane supplier
Utility line, transformer, net metering / interconnection	Electric utility
Gas service (if a main runs along the road)	Gas utility
Dam-safety exemption confirmation	ODNR Division of Water Resources
Mailbox placement	Postmaster
Wildlife pilot and warning signs	Road authority
Bat coordination	ODNR Division of Wildlife / USFWS
Conservation easement (recorded after construction)	Land trust; county recorder
CAUV recoupment and status of eased acres	County auditor

17. Decision Gates: Waiting on the Land, the Energy Model, or Officials

The plan carries a default for each item below until the named source decides it.

17.1 The land decides

Item	Default until known	How it gets decided
Public water or a well	Well on the west side, with drinking-water and sprinkler tanks buried beside it	Utility maps; well logs; health district
Sewer or septic type	Septic on the east shoulder (mound, drip, or aerobic)	Health district soil evaluation, in the same visit as the geotech borings
Natural gas or propane	1,000-gal buried propane tank	Is there a gas main along the road?
Internet	Fiber, with Starlink as a backup	Is fiber at the road?
Oak and limestone yield	Site oak to the pavilion frame first; bought wood and stone cover shortfalls	Forester's cruise of the building area; geotech borings
Every final dimension, including the storm basin's depth	Illustrative 900 × 1,300-ft layout; basin ≥ 2.75 ft average depth	1-ft survey; civil engineer

17.2 The energy model decides

Item	Default until modeled
Solar size	40–50 kW on the house; 10–12 kW on the garage
Battery sizes	LFP 60–80 kWh (capped at 80); NiFe 60–80 kWh if approved
Heat storage tank	Larger than the early 1,000–2,000-gal buffer; mechanical room re-planned for it
Number of geothermal wells	About 15–21 bores
Generator size	UL 2200 standby unit beside the transformer
Electrical service size	NEC 220 load calculation; conduit ready for 800 A

17.3 Officials decide

Item	Default until decided	Who decides
Approval of the nickel-iron batteries	No-go until written approval or an NRTL label	Building official, electrical examiner, fire official
Drive width	24 ft; 26 ft if the dead end counts as over 500 ft	Fire official
Fire hose reach to the house	Sprinkler extension; otherwise the east turf lane	Fire official
How much of the house gets sprinklers	NFPA 13D with extended coverage	Fire official; fire protection engineer
Which code edition to apply under	The current 2019 RCO, applied before any five-year-review rules take effect [A]	Building official
Fire-water reserve	13,000–32,000 gal (my estimate); plan for a 40,000-gal cistern if needed (\$3.6)	Fire official
Tunnel classification and house connection	Rated, gasketed door; tunnels outside the air barrier [A]	Building and fire officials

18. Team, Schedule, and Insurance [F]

When	Who / what	Output
Now , while screening land	Soils (geotechnical) engineer and civil engineer	Parcel screening, borings, slope stability, drainage and septic feasibility
Now	Pre-design talks with the building and fire officials	Fire-water reserve, drive width, hose reach, sprinklers, NiFe path, tunnels, code edition
Under contract	Surveyor (1-ft contours); health district soil evaluation; county auditor (CAUV); forester	Survey; septic type; recoupment credit; oak yield
Once you own a parcel	Architect and energy modeler	Schematic design; energy model sets PV, batteries, heat tank, bores, generator, and service
After schematic design	Structural, MEP, fire-protection, and civil engineers; landscape architect; lighting designer	Tunnels, cistern, and vaults engineered; plans coordinated
First professional estimate	Estimator or construction manager	Budget ceiling; house width; ICF scope
Before permits	Land trust letter of intent; insurance broker	Easement terms; battery-vault quotes

When	Who / what	Output
Phase 1	Contractor	Sequence in §15
After Phase 1	Surveyor; land trust	As-built survey; baseline report; easement recorded
Phase 3	Owner, electrical engineer, building official, insurer	NiFe go/no-go

Insurance. Get quotes on the battery vaults before committing to nickel-iron: first with LFP only, then with NiFe added. Carry builder's risk through construction, and add the tunnels, cistern, and vaults to the property schedule.

19. Century Maintenance Plan

Interval	Tasks
Monthly	Radon log; drains and diverters; pump alarms; battery data; NiFe water level; gate battery; tunnel sump pumps and high-water alarms; CO and radon monitors at the cart bay
Quarterly	ERV filters; cistern screens; generator log; sump inserts; creek walk; ramp drains (top and bottom); tunnel walk for leaks, trays, and lights
Semiannual	Gutters; snow rails before winter; drip ring; dry-hydrant test; vault vents; exercise every cistern valve in the loop; emergency lights and exit signs
Annual	Burrow watch and crest survey; arch and spillway; check spreader lips are level within ± 0.5 in; pool drawdown; late-winter mow; roof walk; sprinklers; radon fan; flexible joint between cistern and loop; pavilion ash pit after the burning season; house-book update
3–5 yr	Exterior wood finish; cistern interior (through the side hatch); lightning-system continuity; well test; propane-tank cathodic-protection test, if propane [A]; NiFe electrolyte (every 7–10 yr)
10–15 yr	Rotate rubber parts; ERV cores; inverters [A]; heat-pump recommissioning; septic pumping; tunnel sump pumps [A]
20–30 yr	Heat-pump compressors (the ground loop stays); LFP and PV replacement [A]
50–75 yr	Standing seam; timber sills; stone re-pointing; slate repairs from attic stock
100+ yr	Re-lay slate on new stainless nails where needed; hand the house book and easement to the next owner

20. Order-of-Magnitude Costs (planning estimates)

These are my own unsourced rough placeholders in 2026 dollars. I did not get to research Ohio construction costs, so replace every line with a professional first estimate before you decide the width. No federal credits are included.

Element	Range
House (17,280 sq ft at about \$600–1,000/sq ft)	\$10–17M
Entrance complex	\$1.5–3.0M
Court, pavilion, cistern, pads, walk	\$1.5–3.0M
Garage + vaults	\$0.8–1.5M

Element	Range
Geothermal, tank, ERVs	\$0.5-0.9M
PV, LFP, generator	\$0.35-0.6M
Water management	\$0.6-1.2M
Septic, well, utilities	\$0.15-0.4M
Landscape	\$0.4-1.0M
Soft costs (12-18%)	\$2-4M
Estate subtotal	≈\$16-28M
Tunnels, Phase 1: house cart bay, main tunnel, forecourt junction, garage tunnel and ramp, full loop, sealed gate stub	\$2.2-5.2M
Tunnels, later: gate tunnel, gate-end chamber, second ramp	\$1.0-2.1M
Total	≈\$19-35M

The tunnel ranges come from a separate rough estimate; the estimator should check whether they carry their own soft costs. A 40,000-gal cistern would push the court line toward its upper end.

Budget ceiling [F]. Set it after the first professional estimate. Hold a 10-15% owner's contingency inside it [R].

Budget levers:

1. **Width:** going from 120 to 110 ft removes 1,440 sq ft, about \$0.9-1.4M at these rates.
2. **ICF scope:** wood framing above the walkout is the fallback, but check its effect on the fire-water reserve (\$3.6) first.
3. **Phasing:** push pads 5-7, the pavilion kitchen, suite fit-out, and the gate tunnel to later phases.
4. **PV:** size it to the metered load.
5. **Cistern as foundation:** saves building a separate structure.
6. **Don't cut:** slate, stainless, drainage, ICF on the walkout, or the tunnel loop. The loop can't be added around the cistern later.

21. Risk Register

Risk	Likelihood / impact	Resolution
Slope instability	Med / Very high	Reject Kope, Eden, and Fairmount sites; borings before purchase; footings to rock
NiFe not approved	Med / Low	NRTL label or alternative-method approval; otherwise use the vault as an expansion room
Battery insurance costly or refused	Med / Med	Quotes before committing to NiFe; LFP-only fallback
Future 80-kWh total cap	Low-Med / Med	Rated separation between vaults; split capacity
Dead end over 500 ft	Med / Low	26-ft drive or a shorter S-bend
Hose reach over 150 ft	Med / Med	Sprinkler extension; east turf lane

Risk	Likelihood / impact	Resolution
Fire reserve takes most of the cistern	Med / Med	Get the number during the land search; ask for a sprinkler reduction; plan a 40,000-gal cistern
Thin storm-storage margin	Med / High	Deeper or wider basin; re-check against Atlas 15
Berm burrows or seepage	Med / High	Clay core; annual watch; armored spillway
Arch outlet blocked	Low / High	Storage already sized with the outlet blocked; spillway
Cistern settlement	Low / Med	Structural tank; flexible joint to the loop; offset tank as fallback
Tunnel leaks or flooding	Low-Med / High	Membrane and drainage board; sumps with duplex pumps and alarms; roofed, drained ramps; storm pipes kept out
Tunnel cost overrun	Med / High	Price it in the first professional estimate; the gate tunnel can wait
Radon or CO entering from the tunnel	Low / Med	Rated, gasketed doors; electric carts only; monitors at the cart bay
Radon	Med / Med	Active system; slight positive pressure; monitors
Chloride misuse	Med / Med	Covenant, easement, signs, grit kept on hand
CAUV loss on eased acres	Med / Med	Auditor check before closing; seller credit on every acre that leaves CAUV
Site oak shortfall	Med / Low	Pavilion frame first; bought heat-treated wood and graded oak cover the rest
Code change mid-design	Med / Med	Code-timing strategy
CGP expires in 2028	High / Low	Calendar the renewal
Cost overrun	High / High	Early estimate; budget ceiling with contingency; width; phasing
Loss of records over time	Med / High	House book with an off-site copy

22. Decision Record

22.1 Committed in Revision B [F]

Decision	Committed	Section
House width	120 ft until the energy model and a professional estimate come back	\$7
Wall system	ICF on all three floors if the budget allows; otherwise wood framing above the walkout	\$7
Phase 1 pads	4 (the other 3 in Phase 2)	\$5, \$15
Brook trickle reservoir	Include	\$10
Conservation easement	Woods, meadow, pool, and outfalls (≈ 16.4 acres); Little Miami Conservancy by default; recorded after construction so the building area can be carved out exactly	\$14
Siding	Site oak once the pavilion frame is covered; bought heat-treated wood covers any shortfall	\$7
Cooking fuel	Induction; no gas in the house	\$7
Budget ceiling	Set after the first professional estimate	\$20

Decision	Committed	Section
Elevator	Commercial limited-use (LU/LA) unit; 25-ft travel covers 22 ft; standard parts	\$7
Window frames	Aluminum-clad wood, with glass units that can be swapped when their seals fail	\$7
Interiors	Plaster walls, cabinets from your own white oak, and a lighting plan	\$7
Floor plans	The \$7 room schedule governs: stair and elevator stacked at 48-72 ft on all levels; library back on the south	\$7
Utility strip	Parts library where cold storage was; wine cellar stays; drinking-water and sprinkler tanks buried outside by the well	\$7, \$9
Cistern	Reaches under all eight posts: 30-40 ft across and shallower; 30,000 gal, or about 40,000 if the fire reserve requires	\$5
Fuel	Natural gas if a main runs along the road; otherwise a 1,000-gal buried propane tank, re-sited clear of buildings, vault vents, creek, septic, and well	\$9
Design loads	115 mph wind, about 20 psf snow, 36-in-deep footings, highest-risk radon zone	\$7
Garage	Walk-in doors on the north and south sides along a 6-ft cart aisle; floor level with the main floor	\$6
Small specs	2-in pattern spacing on bird-safe glass; lit ceilings under covered walks; car-stopping walk posts; no fish in the seasonal pool	\$5, \$7, \$10
Finial	Bronze-colored stainless steel on the cupola (and the pavilion, if used)	\$7, \$12
Tunnels	Loop, main and garage tunnels, garage ramp, five chambers, and sealed gate stub in Phase 1; gate tunnel, chamber, and ramp later	\$13
Insurance	Battery-vault quotes before committing to nickel-iron	\$6, \$18
Team	Soils and civil engineers now; architect and energy modeler once you own a parcel	\$18

22.2 Still open, with defaults applied [R]

Decision	Default applied [R]	Who decides
House width (100/110/120 ft)	Plan carries 120 ft; decide after the energy model and estimate	Owner, architect, estimator
ICF above the walkout	All three floors if the budget allows	Owner, estimator
Budget ceiling amount	Set at the first estimate, with a 10-15% contingency inside it	Owner
Storm margin	6.84 in; re-check against Atlas 15	Civil engineer
Cistern as foundation vs. offset tank	Foundation, if bearing is uniform	Structural engineer + geotech
Cistern size	30,000 gal; 40,000 if the fire reserve requires it	Fire official; owner
Sprinkler scope	NFPA 13D with extended coverage	Fire protection engineer + fire official
Code-timing strategy	Apply under the current 2019 RCO before any five-year-review rules take effect; confirm the governing edition at application [A]	Architect + building official
NiFe go/no-go	No-go until written approval, acceptable insurance, and 1-2 years of metered data	Owner, electrical engineer, building official
Drive width	26 ft if the dead end is over 500 ft	Fire official
Hose reach	Sprinkler extension, otherwise the east lane	Fire official

Decision	Default applied [R]	Who decides
Basin depth	At least 2.75 ft average below the storage line	Civil engineer
House PV size	40–50 kW	Energy model
Tunnel section	About 10 ft wide × 9 ft clear, precast box	Structural + civil engineers
Propane tank location	Northeast hilltop, if there's no gas main	Civil engineer, supplier, fire official
Land trust	Little Miami Conservancy if the parcel is in its watershed	Owner; land trust

Caveats

- **Costs:** unsourced placeholders, including the tunnel ranges.
- **Rainfall:** the Atlas 14 values come from a point query at Lebanon. The actual site, and Atlas 15, may change the design depth.
- **Code review:** the RCO five-year review is still pending.
- **Assumptions:** items marked [A] need confirmation from the named authority.
- **Floor plans:** I rebuilt the \$7 schedule on the 120-ft grid from the rooms in the Berm Ring draft, following the fixes above. If you have the original room schedules, compare them room by room; where they differ on a room's size or neighbor, the originals govern, as long as the stair core stays stacked and the library faces south.
- **Process:** the earlier draft's automated enrichment pass never ran, because the draft was over the tool's size limit on every attempt. The figures carried over come from the official sources checked directly in that draft, or from the targeted rainfall lookup. Its source links did not survive printing to PDF. The new checks in this revision are listed under Sources.

Appendix A. Where Each Revision Item Landed

Revision item	Where it is now
Needs committed: house width; ICF; 4 Phase 1 pads and the brook reservoir; easement acres, land trust, and timing; siding; cooking fuel; budget ceiling	\$22.1, with detail in \$5, \$7, \$10, \$14, \$20
Waiting on the land: water, sewer or septic, gas or propane, fiber with Starlink backup, oak and limestone yield, final dimensions and basin depth	\$17.1
Waiting on the energy model: solar, batteries, heat tank, geothermal wells, generator, service size	\$17.2
Waiting on officials: NiFe approval, drive width, hose reach, sprinkler extent, code edition	\$17.3
Fire water: 13,000–32,000 gal estimate; sprinkler reduction; ~40,000-gal tank	\$3.6, \$5, \$17.3
Never specified: elevator, window frames, interiors, insurance, team and schedule	\$7, \$6, \$18
Floor plans (stair/elevator stacking, library south, original schedules govern)	\$7
Utility strip (parts library, wine cellar, tanks by the well)	\$7, \$9
Cistern 30–40 ft across and shallower	\$5
Fuel and propane re-siting	\$9

Revision item	Where it is now
Design loads	\$7
Garage doors , cart aisle, floor level	\$6
Small specs (bird-safe glass, lit ceilings, car-stopping posts, no fish)	\$5, \$7, \$10
Copper finial → bronze-colored stainless	\$7, \$12
Loop around the cistern (connections, tank pipes, chimney, posts and exits)	\$13.2, \$5
Turnaround spots (chambers, cart bay, pull-outs)	\$13.3
Cart entrances (garage ramp now, gate ramp later, size, weather)	\$13.4, \$6
Utilities in the tunnels	\$13.5, \$9
Serviced from the tunnels	\$13.6
Stays out, and why	\$13.7
Cost and phasing of the tunnels	\$13.9, \$15, \$20

Sources (checks added in Revision B)

- Three Valley Conservation Trust, service area and accreditation: [Farmland Information Center](#)
- Tecumseh Land Trust, service area and accreditation: [Land Trust Alliance](#), [Farmland Information Center](#)
- Little Miami Conservancy, service area and the 2024 First Creek easement: [HabitatCAN](#), [Little Miami Conservancy](#)
- CAUV woodland must be devoted to commercial timber production: [TimberTax: Ohio property tax](#)
- CAUV conservation land valued at the lowest soil value (HB 49, 2017): [OSU Farm Office](#)